Methods and Results

Methods

Character based methods were to create a phylogenetic tree using the software RAxML; This software utilizes maximum likelihood techniques to establish the most likely phylogenetic tree of a multiple sequence alignment. Using the primate data that was provided, it could be plugged into a “guey” that would run a comparison and use boot strapping to provide a tree. This guey also provides the code that would be used in a computer terminal using normal RaxML software.

Results

Using these methods, a well-refined tree is produced. Making use of this tree an individual can clearly visualize the evolutionary relationships of the taxa, in this example the primates. In this tree, it showed more of grouping between two groups with boot strapping data to support this. It also revealed that lemurs and Tarsius formed an out group.

Relevance

This process and results reveal the evolutionary relationships between the chosen multiple sequence alignments samples. This information can be insightful to professionals in many fields. For example, this can be used to revise taxonomy, calculate diversification, study biogeography, etc. Such as understanding homologous genes when understanding functions that may have been passed down.